

Model Evaluation Report — RAG + LLM (Infant & Young Child Feeding)

1. Objective

Assess end-to-end performance of the RAG pipeline on a 60-question benchmark covering six query categories (fact_check, summarization, noisy_grammar, multilingual, semantic_paraphrase, on_topic_outside_context). Focus areas: retrieval quality, answer grounding, relevance, and adherence to the 140-character response constraint.

2. Evaluation Setup (concise)

- **Data source:** WHO IYCF fact sheet–derived questions (10 per category, total n=60).
- **Pipeline:** FAISS dense retrieval → prompt templating → LLM generation (≤140 chars).
- **Recorded metrics**
 - **Relevance:** qa_cosine (answer↔question)
 - **Retrieval match:** q_ctx_cosine (question↔retrieved context)
 - **Grounding:** a_ctx_cosine (answer↔context), rougeL_recall_a_ctx (answer coverage from context), answer_novelty_vs_ctx (lower = safer)
 - **Lexical checks:** jaccard_q_ctx, jaccard_a_ctx
 - **Format:** response_length, within_char_cap

3. Overall Results (n=60)

Run summary:

qa_cosine **0.4378**, q_ctx_cosine **0.3628**, a_ctx_cosine **0.5219**, jaccard_q_ctx **0.0119**, jaccard_a_ctx **0.0697**, rougeL_recall_a_ctx **0.8472**, answer_novelty_vs_ctx **0.1510**, within_char_cap **100.0%**, avg_response_length **105.0**

3.1 Key observations

- **Grounding strength is high.**
rougeL_recall_a_ctx at **0.8472** combined with **low** answer_novelty_vs_ctx (**0.1510**) indicates the generated answers are well supported by retrieved evidence and exhibit limited hallucination.
- **Retrieval is the primary bottleneck.**
q_ctx_cosine at **0.3628** trails a_ctx_cosine (**0.5219**). This gap suggests that once context is provided, the model aligns well with it, but the retrieved context is not consistently optimal for the question.

- **Answer relevance is moderate.**

qa_cosine at **0.4378** indicates answers are broadly on topic but not consistently crisp across all query forms.

- **Excellent output control.**

100% adherence to the ≤ 140 -character constraint with an average length of **105** characters reflects good prompt and post-processing discipline.

3.2 Interpretation of supporting metrics

- **Lexical overlaps** (jaccard_q_ctx **0.0119**, jaccard_a_ctx **0.0697**) are low, which is typical for short, paraphrased outputs. The high ROUGE-L mitigates concern here.
- **Safety proxy** (answer_novelty_vs_ctx **0.1510**) is comfortably low, consistent with grounded responses.

4. Category-Level Insights

The provided summary contains overall metrics; per-category aggregates (means/medians/IQRs) are produced by the analysis notebook. The statements below reflect the comparative patterns visible in the per-category box plots and mean charts generated for this run.

- **Retrieval match (q_ctx_cosine)**
Categories with language noise and non-English inputs exhibit **lower central tendency** and **wider dispersion** (noisy_grammar, multilingual). Fact-formulated items (fact_check) show the **highest** central tendency and **tightest** spread. Semantic paraphrases trend between these groups.
- **Grounding (a_ctx_cosine, rougeL_recall_a_ctx, answer_novelty_vs_ctx)**
Grounding remains **consistently strong** across categories; even where retrieval alignment dips, rougeL_recall_a_ctx remains high and answer_novelty_vs_ctx remains low, indicating that when useful evidence is retrieved, the generator stays anchored to it.
- **Answer relevance (qa_cosine)**
Fact-focused and direct questions outperform broad summarization prompts. Multilingual and noisy_grammar categories show **lower medians** and **longer tails**, reflecting sensitivity to query formulation.
- **Constraint adherence**
All categories meet the character limit; distributions cluster below the cap, suggesting no category systematically requires more budget.

Recommendation for documentation: include the per-category table exported from the notebook (category_summary.csv) and reference selected plots (e.g., “Figure 2: q_ctx_cosine by category”) to anchor the qualitative comparisons above.

5. Action Plan (prioritized)

A. Retrieval improvements (largest impact)

1. Hybrid retrieval (dense + lexical)

Combine FAISS with BM25 (e.g., ensemble weights 0.6 / 0.4). Expect increased q_ctx_cosine, especially for paraphrase and noisy queries.

2. MMR diversification

Use fetch_k≈20, k≈5–8, lambda_mult≈0.2–0.5 to reduce redundancy and surface diverse but relevant chunks.

3. Cross-encoder re-ranking

Re-score top-20 candidates with a lightweight cross-encoder (e.g., ms-marco-MiniLM-L-6-v2), keep top-5. This typically lifts both q_ctx_cosine and downstream qa_cosine.

4. Chunking adjustments

Reduce chunk size to **300–400 chars** with **80–120 overlap** to improve hit-rate on atomic facts.

B. Query normalization and multilingual handling

5. Lightweight normalization for noisy inputs

Lowercasing, punctuation cleanup, and common misspell corrections prior to retrieval to stabilize q_ctx_cosine in noisy_grammar.

6. Multilingual bridge

Either translate non-English queries to English before retrieval or swap to a **multilingual embedding model** for the vector store and query encoder.

C. Generation controls (targeted)

7. Prompt nudge for directness

Add a short directive (e.g., “Answer the question directly; avoid unrelated details.”) to lift qa_cosine without harming grounding.

8. Category-specific caps (optional)

Maintain 140 chars overall; consider a **120-char** cap for summarization items if responses cluster near the limit without adding content.

D. Instrumentation and evaluation

9. Return exact prompt context (optional)

Add an `include_context=True` option to the pipeline to store the **exact** context string used. This aligns metrics with the true prompt input and removes reconstruction uncertainty.

10. Category KPIs

- *fact_check*: add exact-match or keyword checks (“6 months”, “within 1 hour”, “2 years or beyond”).
- *on_topic_outside_context*: track correct denials (“Not in knowledge base.”) as a success rate.

11. Quality gates

Introduce thresholds in CI (tune post-next run), e.g., $\text{avg_q_ctx_cosine} \geq 0.50$, $\text{avg_a_ctx_cosine} \geq 0.60$, $\text{avg_qa_cosine} \geq 0.55$, $\text{answer_novelty_vs_ctx} \leq 0.12$, $\text{within_char_cap} = 100\%$.

6. Targets for the next iteration

- $\text{q_ctx_cosine} \geq \mathbf{0.50}$ (from 0.3628)
- $\text{a_ctx_cosine} \geq \mathbf{0.60}$ (from 0.5219)
- $\text{qa_cosine} \geq \mathbf{0.55}$ (from 0.4378)
- Maintain $\text{rougeL_recall_a_ctx} \geq \mathbf{0.80}$ and $\text{answer_novelty_vs_ctx} \leq \mathbf{0.12}$
- Maintain **100%** within-char-cap

7. Risks and mitigations

- **Copy-without-answering risk** (high ROUGE-L but low qa_cosine): mitigate with prompt nudges and re-ranking to ensure retrieved evidence aligns with the actual question.
- **Multilingual sensitivity**: where translation is unacceptable (privacy/offline), prioritize multilingual embeddings; otherwise, apply on-device translation before retrieval.
- **Over-compression risk**: if a shorter cap degrades interpretability for fact_check items, keep 140 for those while testing 120 elsewhere.

Conclusion. The evaluation indicates **strong grounding** and **excellent format control**, with the principal improvement area in **retrieval alignment**. Implementing hybrid retrieval, MMR, and cross-encoder re-ranking—along with light query normalization and

multilingual handling—is expected to materially raise q_ctx_cosine and qa_cosine while preserving grounding and brevity.